# JIAYI WANG

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## RESEARCH INTERESTS

- Computer architecture for AI and general-purpose workloads (CPU/GPGPU/accelerators).
- Reconfigurable and spatial/dataflow computing architectures (FPGA/CGRA).
- Memory systems, cross-stack co-design, and multimodal AI systems for real-world applications.

## EDUCATION

**University of Washington, Seattle** Sep 2023 – June 2028 (expected)
Ph.D. in Computer Engineering
GPA: 4.00
Focus: Computer Architecture, reconfigurable architecture, AI systems, digital VLSI
Advisors: Prof. Ang Li and Prof. C.-J. Richard Shi

**University of Washington, Seattle** Sep 2020 – Aug 2022
M.S. in Electrical Engineering
GPA: 4.00
Focus: Computer architecture, digital VLSI

**Shanghai Jiao Tong University** Sep 2016 – Jun 2020
B.S. in Electrical Engineering
GPA: 3.70
Focus: Power electronics, embedded systems

**University of California, Los Angeles** Summer 2019
Summer School, Computer Science
GPA: 4.00

## FIRST-AUTHOR PUBLICATIONS AND WORKSHOPS

[1] **Jiayi Wang**, Ang Da Lu, Zhichen Zeng, and Ang Li,
"*DICE: Enabling Efficient General-Purpose SIMT Execution with Statically Scheduled Coarse-Grained Reconfigurable Arrays*,"
in **The 53rd Annual International Symposium on Computer Architecture (ISCA 2026)**, Raleigh, NC, USA, Jun. 27–Jul. 1, 2026. (Acceptance rate: 19.1%.) [arXiv:2605.05496]

[2] **Jiayi Wang**, Maohua Nie, Sin-Chen Lin, C.-J. Richard Shi, and Ang Li,
"*TransDot: An Area-Efficient Reconfigurable Floating-Point Unit for Trans-Precision Dot-Product Accumulation for FPGA AI Engines*,"
in **The 34th IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM 2026)**, Atlanta, GA, USA, May 13–16, 2026. (Acceptance rate: 24.6%.) [IEEE Xplore]

[3] **Jiayi Wang**, Ang Da Lu, and Ang Li,
"*DICE: Efficient Thread-Pipelined SIMT Execution on a Statically Scheduled Reconfigurable Spatial Fabric*,"
**Presented at 18th Workshop on General Purpose Processing Using GPU (GPGPU 2026)**, held in conjunction with ASPLOS 2026, Pittsburgh, PA, USA, Mar. 22–23, 2026.

[4] **Jiayi Wang**, Chenyi Wang, and Ang Li,
"*piPE-SA: Enabling Deeply Pipelined Processing Elements in Systolic Arrays*,"
in **Energy Efficient Machine Learning and Cognitive Computing (EMC2-11) Workshop**, held in conjunction with ASPLOS 2026, Pittsburgh, PA, USA, Mar. 22, 2026.

[5] **Jiayi Wang**, Yiwei Yang, Jiajin Wu, Zheng Zeng, Di Lu, and Lian Lian,
"*Hybrid Aerial-Aquatic Vehicle for Large-Scale High-Spatial-Resolution Marine Observation*,"
in **OCEANS 2019 Marseille**, Jun. 2019, pp. 1–7. doi: 10.1109/OCEANSE.2019.8867402.

## COLLABORATIVE PUBLICATIONS

[6] Ryan H. Lee, Kate L. Tseng, Te Min Yu, Andrew Pan, **Jiayi Wang**, James Rosenthal, Kevin J. Ho, Ang Li, and Matthew Reynolds,
"*All-Digital Bluetooth Low Energy (BLE) Backscatter ASIC Using Standard I/O Pad Drivers in 180 nm CMOS*,"
in **Proceedings of the 2026 IEEE International Conference on RFID (RFID 2026)**.

[7] Jingqun Zhang, Weihang Li, Maohua Nie, Yung-Jen Cheng, **Jiayi Wang**, Shwet Chitnis, and Ang Li,
"*CacheFlex: Explicitly Control What You Need in Your Cache*,"
**Presented at Energy Efficient Machine Learning and Cognitive Computing (EMC2-11) Workshop**, held in conjunction with ASPLOS 2026, Pittsburgh, PA, USA, Mar. 22, 2026.

[8] Ang Da Lu, **Jiayi Wang**, and Ang Li,
"*STEP: Spatially Threaded Execution Pipeline*,"
in **Proceedings of the 6th Workshop on Languages, Tools, and Techniques for Accelerator Design (LATTE 2026)**, held in conjunction with ASPLOS 2026, Pittsburgh, PA, USA, Mar. 22–23, 2026. [PDF]

[9] Shwet Chitnis, Fergus Xu, Ayush Kulkarni, **Jiayi Wang**, Jingqun Zhang, Arjun Raje, and Ang Li,
"*Precision-Aware Communication in CGRAs*,"
in **The 34th IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM 2026)**.

[10] Jingqun Zhang, Maohua Nie, **Jiayi Wang**, Weihang Li, Rishi Sappidi, Shwet Chitnis, and Ang Li,
"*CacheFlex: Direct Software-Managed Access to Higher-Level Cache for Scalable Vector Support*,"
in **The 59th IEEE/ACM International Symposium on Microarchitecture (MICRO 2026)**. (Acceptance rate: 25.3%.)

## SUBMITTED AND ONGOING MANUSCRIPTS

[11] Zhichen Zeng, Xichong Zhang, Junpan Wu, Yifei Zuo, Chi-Chih Chang, **Jiayi Wang**, Maohua Nie, Ji Liu, Ang Li, and Banghua Zhu,
"*Step-dLLM: Adaptive Step-aware Sparse Attention for Efficient Diffusion LLM Inference*,"
**Under review at NeurIPS 2026 (Conference Track)**.

[12] Zhichen Zeng, Chi-Chih Chang, **Jiayi Wang**, Zezhou Wang, Ningxin Zheng, Zheng Zhong, Cesar Andres Stuardo Moraga, Dongyang Wang, Mohamed S. Abdelfattah, Haibin Lin, Banghua Zhu, Ang Li, and Ziheng Jiang,
"*DisagMoE: Computation-Communication Overlapped MoE Training via Disaggregated AF-Pipe Parallelism*,"
**Under review at NeurIPS 2026 (Conference Track)**.

[13] Maohua Nie, Jiang Zhu, Jingqun Zhang, Zhichen Zeng, **Jiayi Wang**, Sibo Zhang, Jialin Wang, and C.-J. Richard Shi,
"*HierSVA: A Synthesis Pipeline, Dataset, and Benchmark for LLM-Driven Hierarchical Hardware Formal Verification*,"
**Under review at NeurIPS 2026 (Datasets and Benchmarks Track)**.

## PATENTS

[1] Ang Li and **Jiayi Wang**,
"*DICE: Enabling Efficient General-Purpose SIMT Execution with Coarse-Grained Reconfigurable Arrays*,"
**U.S. Provisional Patent Application 64/012,051**, filed Mar. 20, 2026.

[2] Zheng Zeng, **Jiayi Wang**, Yiwei Yang, Jiajin Wu, Lian Lian, and Di Lu,
"*An Aerial-Aquatic Vehicle Combining a Fixed-Wing Aircraft and a Glider*,"
**Chinese Invention Patent CN 109204812 B**, granted Nov. 17, 2020.

## SELECTED RESEARCH EXPERIENCE

### RTA: Reconfigurable Tensor Arrays for Multiprecision AI and General-Purpose Computing

*UW $P^N$ Computer Engineering Lab (PI: Prof. Ang Li)* — *Aug 2025 – Present*

- Designed a twisted-torus systolic array that eliminates shifter registers and reduces latency relative to conventional systolic arrays.
- Developed a reconfigurable processing element supporting general-purpose vector operations beyond matrix multiplication, improving utilization across diverse AI workloads.
- Developed **TransDot**, a reconfigurable floating-point unit supporting FP4/FP8/FP16/FP32 fused multiply-add, subword SIMD execution, and dot-product accumulation in FP16/FP32.

### VISTA: Video Interaction Scoring and Teaching Assistant

*SABOL Vision Lab, Cultivate Learning (PI: Prof. Gail E. Joseph)* — *Feb 2026 – Present*

- Building a multimodal AI system for early-childhood classrooms that enables automated interaction-quality scoring and low-latency teacher coaching.
- Developed **VISTA-QEI**, a video analysis pipeline for scoring teacher-child interactions using the QEI framework and generating evidence-based coaching reports.
- Designed **VISTA-RT**, a real-time coaching system with live video/audio analysis, rolling context memory, and prioritized teacher prompts delivered through wearable interfaces.
- Implemented working demos and evaluation tools, and presented prototype results to interdisciplinary collaborators across AI and early learning.

### DICE: A CUDA-Compatible CGRA-Based Energy-Efficient GPGPU

*UW $P^N$ Computer Engineering Lab (PI: Prof. Ang Li)* — *Oct 2024 – Present*

- Proposed a novel GPGPU architecture that replaces the SIMD backend with statically scheduled CGRAs to reduce register-file pressure and control overhead.
- Developed a Temporal Memory Coalescing Unit (TMCU) for pipelined memory coalescing and introduced swizzled thread-bank mapping to reduce register-file bank conflicts.
- Extended GPGPU-Sim/GPUWattch and built a *p-graph* compiler for performance and power modeling, evaluated on GPU-Rodinia benchmarks.
- Led an 8-member RTL team to build an RTL implementation and FPGA prototype for architectural validation.
- Achieved **1.90× energy efficiency**, **42% lower power**, and **1.16× performance** relative to NVIDIA Turing SMs.

### DORA: A Python-Based CGRA Prototyping Framework

*UW $P^N$ Computer Engineering Lab (PI: Prof. Ang Li)* — *Mar 2024 – Present*

- Built a highly parameterized CGRA RTL generation framework enabling rapid customization of processing elements, routing networks, and control flows.
- Accelerated architecture exploration and prototyping for CGRA-based research projects through reusable Python-based hardware generation infrastructure.

### CPHY: DDR/UCIe-Compatible PHY and LPDDR4x Memory Controller

*UW Silicon Systems Research Lab (PI: Prof. C.-J. Richard Shi)* — *Mar 2022 – Dec 2023*

- Led RTL design, mixed-signal integration, and verification of an LPDDR4/4x memory controller and DDR/UCIe-compatible PHY.
- Performed physical design in TSMC N28, including AMS integration.
- Integrated the memory controller and PHY with a RISC-V CPU and a deep-learning accelerator.

## INDUSTRY EXPERIENCE

**NVIDIA Corporation** Summer 2026
*Compute Performance Intern*

- Working on GPU acceleration of Cadence EDA tools.

**Qualcomm Technologies** Jun 2024 – Sep 2024
*Research Intern, GPGPU*

- Investigated architectural optimizations in commercial GPGPU/NPU platforms to accelerate sparse matrix multiplication.
- Explored low-level sparse implementation flows in TensorRT, PyTorch, and NVIDIA CUTLASS using NVIDIA 2:4 Sparse Tensor Cores.
- Benchmarked and analyzed sparse Tensor Core performance across multiple models, including ResNet, MobileNet, and BERT.

**Orora Design Technologies** Oct 2022 – Dec 2023
*Technical Staff*

- Developed C++ automation tools for automatic SystemVerilog model generation of analog circuits.

**Intel Corporation** Mar 2021 – Sep 2021
*Technical Enablement Engineer Intern*

- Debugged and verified DDR and PCIe high-speed interfaces on IA-32 platform products.

**Signify (Philips Lighting)** Aug 2019 – Feb 2020
*Hardware Engineer Intern*

- Worked on a dual-mode self-adaptive LED/HID driver circuit for the EU market.

## TEACHING, MENTORING, AND SERVICE

**University of Washington, Seattle** Jan 2024 – Present
*Graduate Teaching Assistant*

- Courses supported: EE478 Chip Testing, EE478 VLSI Capstone, EE477/525 VLSI-II, EE476 VLSI-I, and EE271 Digital Circuits.
- Led and mentored 5 TSMC180 tapeout projects.
- Developed a complete ASIC tapeout flow for TSMC 180 nm technology using the `Hammer` flow with custom TCL plugins.
- Implemented and maintained synthesis (Genus), place-and-route (Innovus), STA (Tempus), DRC/LVS (Calibre), padframe generation, and hierarchical flows with extracted timing models.

**University of Washington, Seattle** Winter 2025
*M.S. Admissions Triage Committee*

- Reviewed master's applicant materials and provided evaluation feedback.

**UW Semiconductor Society** Dec 2025 – Present
*Technical Member*

- Contributed to student community activities related to semiconductor design and systems research.

## EXPERTISE AND SKILLS

- **Architecture and Modeling:** Strong background in AI hardware and CPU/GPGPU/FPGA/CGRA architecture; experienced with GPGPU-Sim/GPUWattch (Accel-Sim/AccelWattch) and gem5.

- **ASIC Design:** End-to-end ASIC design, tapeout, and mentoring experience in TSMC 180 nm and N28, including RTL design, verification, synthesis, place-and-route, STA, and signoff (DRC/LVS/PEX).
- **EDA Tools:** Cadence (Xcelium, Genus, Innovus, Tempus), Synopsys, and Mentor/Calibre.
- **Programming:** CUDA, Python, C++, and SystemVerilog; strong debugging skills and experience working with complex codebases.